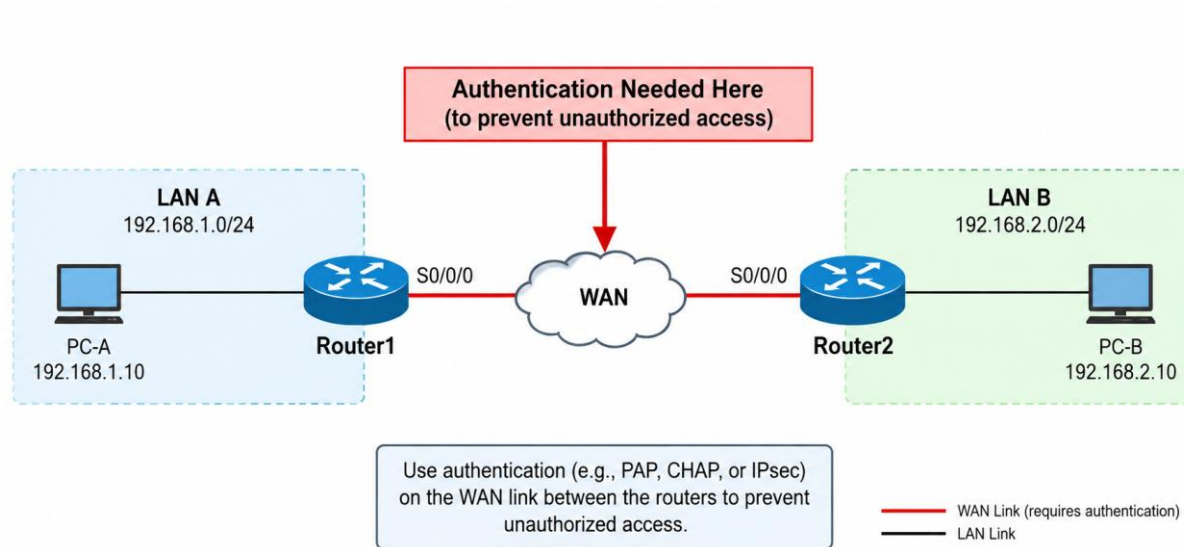


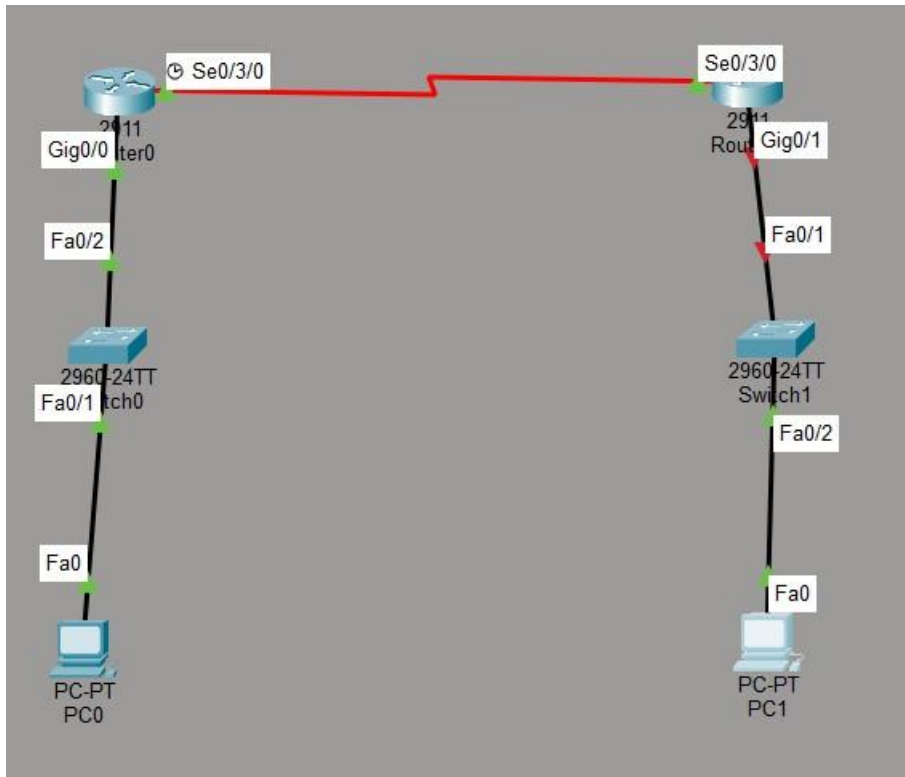
task-1



Task-2

**Q-Configure a sample HDLC WAN connection between two routers using Packet Tracer or any network simulator, and explain why HDLC does not provide authentication.

Hint: Focus on the commands used and what is missing in terms of security.**



Router 0

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int s0/3/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#en
Router(config-if)#encapsulation hdlc
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#no shut
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#
```

Router 1

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

Router(config-if)#exit
Router(config)#int s0/3/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#en
Router(config-if)#encapsulation hdlc
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
%Invalid next hop address (it's this router)
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

```

Router#show interfaces serial 0/3/0
Serial0/3/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.0.0.2/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations 0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions
  DCD=up DSR=up DTR=up RTS=up CTS=up

```

Why doesn't HDLC provide authentication?

HDLC (High-Level Data Link Control) is a Layer 2 data-link protocol. Its basic function is to encapsulate and transport frames over a WAN link.

HDLC does **not** provide a built-in username/password authentication mechanism.

For example, notice our configuration:

```
interface serial 0/3/0  
ip address 10.0.0.1 255.255.255.252 encapsulation hdlc
```

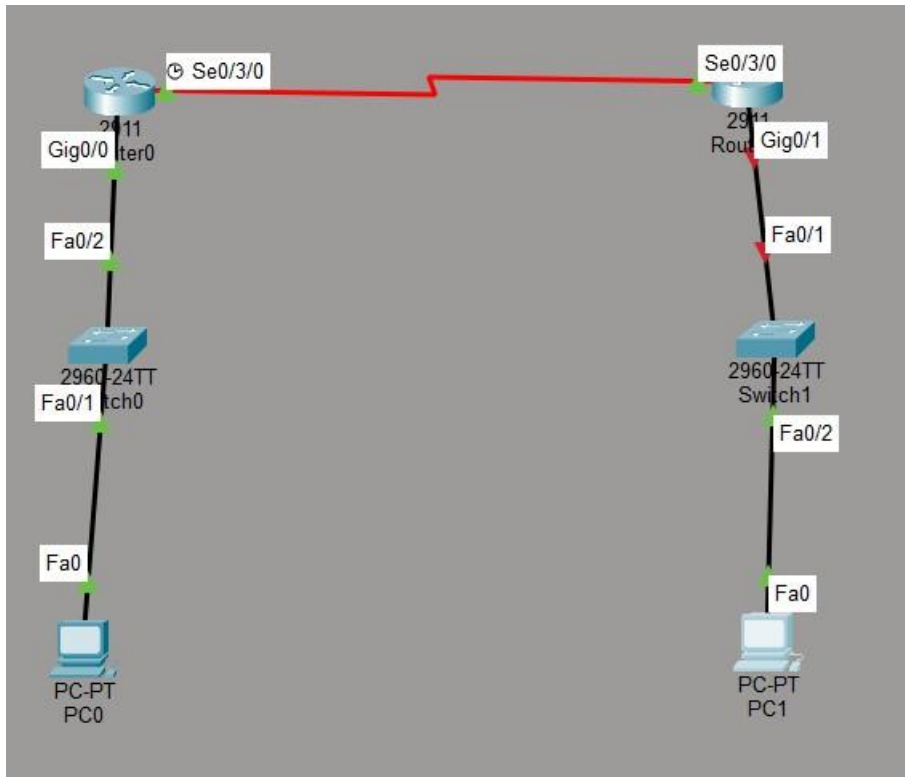
There is **no command such as:**

```
username password authentication
```

HDLC simply establishes the data-link communication; it does not authenticate the identity of the peer router.

Task -3

Q-Set up a PPP connection between two routers in Packet Tracer and enable PAP (Password Authentication Protocol) authentication. Show the configuration commands and explain what happens if the password is incorrect.



Router 0

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#username R2 password hello1
R1(config)#int s 0/3/0
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#en
R1(config-if)#encapsulation ppp
R1(config-if)#ppp authentication pap
R1(config-if)#ppp pap sent-username R1 password hello1
R1(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
R1(config-if)#exit
R1(config)#do wr
Building configuration...
[OK]
R1(config)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

R1(config)#exit

```

Router 1

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#username R1 password hello1
R2(config)#int s0/3/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#en
R2(config-if)#encapsulation ppp
R2(config-if)#ppp authentication pap
R2(config-if)#ppp pap sent-username R2 password hello1
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

R2(config-if)#exit
R2(config)#

```

```

R1#ping 10.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/9 ms

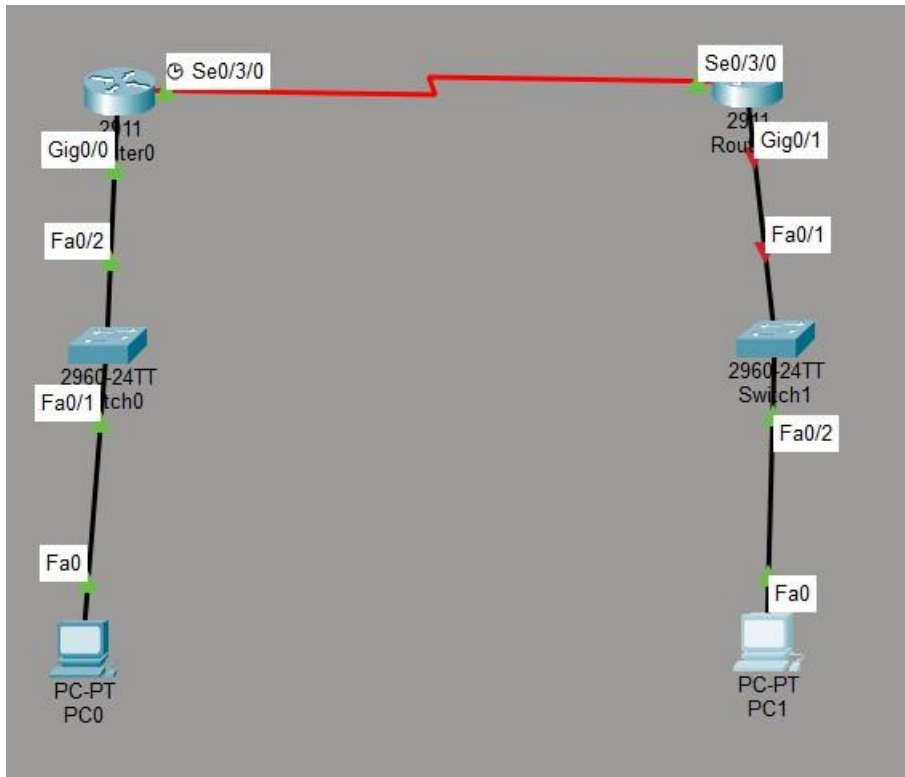
R1#show interfaces serial 0/3/0
Serial0/3/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.0.0.1/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation PPP, loopback not set, keepalive set (10 sec)
  LCP Open
  Open: IPCP, CDPCP
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations 0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
  5 minute input rate 5 bits/sec, 0 packets/sec
  5 minute output rate 5 bits/sec, 0 packets/sec
    5 packets input, 640 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    5 packets output, 640 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions
  DCD=up DSR=up DTR=up RTS=up CTS=up

R1#

```

Task-4

Q-Set up a PPP connection between two routers in Packet Tracer and enable PAP (Password Authentication Protocol) authentication. Show the configuration commands and explain what happens if the password is incorrect.



Router 0

```
R1(config)#interface serial 0/3/0
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#encapsulation ppp
R1(config-if)#ppp authentication chap
R1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to down

R1(config-if)#no shut
R1(config-if)#exit
R1(config)#do wr
Building configuration...
[OK]
R1(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 1

```
R2(config)#interface serial 0/3/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#encapsulation ppp
R2(config-if)#ppp authentication chap
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
```

Ping & Verify

```

R1#ping 10.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/11/25 ms

R1#show interfaces serial 0/3/0
Serial0/3/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.0.0.1/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation PPP, loopback not set, keepalive not set
  LCP Open
  Open: IPCP, CDPCP
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
     Conversations  0/0/256 (active/max active/max total)
     Reserved Conversations 0/0 (allocated/max allocated)
     Available Bandwidth 1158 kilobits/sec
  5 minute input rate 5 bits/sec, 0 packets/sec
  5 minute output rate 5 bits/sec, 0 packets/sec
    10 packets input, 1280 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    10 packets output, 1280 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions
  DCD=up DSR=up DTR=up RTS=up CTS=up

R1#
R1#

```

Task-5

Scenario 1: ISP Subscriber Connections over PPPoE (DSL & Fiber Broadband)

- **Protocols Used:** CHAP / MS-CHAPv2 (or PAP in legacy deployments) running over PPPoE (*Point-to-Point Protocol over Ethernet*).

Real-World Context: When a residential or small-business router powers on and connects to an Internet Service Provider (ISP) via DSL or Fiber-to-the-Home (FTTH), the router initiates a PPPoE session. The ISP's Broadband Network Gateway (BNG) or RADIUS server prompts the router to authenticate using CHAP or MS-CHAP before granting WAN access.

Why Authentication is Critical:

Access Control & Revenue Protection: It prevents unauthorized users from physically plugging a device into an active line or fiber splitter to siphon free internet service without a paid subscription.

Bandwidth Provisioning & Billing: It identifies the exact subscriber account, enabling the ISP to dynamically assign public IP addresses, enforce contracted speed tiers (e.g., 100 Mbps vs. 1 Gbps), and track data usage for billing purposes.

Scenario 2: Remote-Access Enterprise VPNs (L2TP/IPsec & RADIUS Networks)

- **Protocols Used:** MS-CHAPv2 (often encapsulated inside EAP-MSCHAPv2 or protected L2TP/IPsec tunnels).

Real-World Context: When remote employees log into a company's corporate network from home or public Wi-Fi, the enterprise VPN gateway prompts for their domain credentials. The gateway passes these credentials via a RADIUS server to an internal directory service (such as Microsoft Active Directory) using MS-CHAPv2 to verify identity.

Why Authentication is Critical:

Perimeter Defense & Data Protection: It blocks unauthorized external users on the public internet from infiltrating internal networks, databases, and confidential file servers.

Accountability & Role-Based Access Control (RBAC): It ensures that incoming encrypted tunnel traffic is mapped to a verified individual identity, allowing network administrators to restrict access to specific subnets and maintain detailed security audit logs for compliance.

